

Regional Favoritism in Later Periods: Nightlights and Pollution in Leader Birth Regions

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We study how strongly the canonical birth-region favoritism result in nightlights appears in later annual windows. We combine the Political Leaders' Affiliation Database, ADM2-level nightlights, ACAG surface NO₂, ACAG PM_{2.5}, V-Dem democracy scores, and GADM administrative boundaries in a global district-year panel with region fixed effects and country-year fixed effects. The long DMSP replication window recovers the classic result: the lag-0 coefficient is 0.0128 in 1992–2013 ($p = 0.030$) and remains positive in nearby historical windows. In the post-2005 annual windows, the estimates are much smaller and statistically uninformative: 0.0017 in DMSP 2005–2013 ($p = 0.768$) and 0.0004 in harmonized 2005–2019 ($p = 0.966$). A targeted search across geography, tenure, autocracy, luminosity headroom, and pollution-support subsamples shows that Africa and Sub-Saharan Africa drive the historical result, but no post-2005 subsample restores a precise activity effect. Pollution outcomes test one explanation for attenuation: favoritism may have shifted from the quantity to the composition of growth. The estimates do not support that interpretation. Pooled NO₂ and pollution-intensity estimates are positive, though less robust under country clustering, and PM_{2.5} results are also not negative. The main contribution is to document broad post-2005 attenuation and to show that cleaner birth-region growth does not explain it.

regional favoritism | political economy | nighttime lights | air pollution | autocracy

Political leaders often direct resources toward their home regions, a pattern documented in cross-country evidence using subnational nightlights (1). This paper studies how strongly that result appears in later annual windows. The post-2005 years are especially useful for that question because they overlap with harmonized nightlights and global pollution measures. They are also exactly where the canonical birth-region activity effect becomes much harder to detect.

This paper documents that attenuation and asks what it implies. In the long DMSP window, the lag-0 birth-region coefficient is positive and statistically significant. In 2005–2013, still within DMSP, and in harmonized 2005–2019, the point estimates are much smaller and statistically uninformative. The attenuation is not confined to one endpoint choice. It also survives targeted cuts by geography, leader tenure, autocracy, luminosity headroom, and pollution-support samples.

Pollution outcomes enter as a diagnostic for one possible explanation. If leaders still favor birth regions after 2005, but do so through the composition rather than the quantity of growth, then birth regions may receive cleaner activity even when nightlights do not rise. We define “green favoritism” as lower pollution per unit of proxied economic activity in leader birth regions. This possibility matters for environmental inequality as well as distributive politics (2). The pollution evidence does not support it: NO₂ and PM_{2.5} estimates are zero to positive rather than negative.

The contribution relative to recent replication work is therefore diagnostic. Recent work with expanded birthplace data and harmonized nightlights also finds a positive pooled favoritism effect in extended samples (3, 4). By contrast, we focus on the annual post-2005 windows that overlap with our pollution outcomes, test theory-favored subsamples where the nightlights reduced form should be strongest, and evaluate whether a shift toward cleaner growth can explain the weaker activity effect. The result is not that the historical finding disappears in all data. It is that in the later annual windows we study, the classic birth-region nightlights effect is much weaker and statistically uninformative.

1. Data and Empirical Design

The analysis combines five core inputs. First, the Political Leaders' Affiliation Database (PLAD) provides leader identity, spell timing, and subnational birthplace information used to code treatment (5). Second, nightlights provide the economic-activity proxy. We use a replication-style DMSP ADM2 panel for 1992–2013 and a harmonized DMSP-to-VIIRS panel for 2005–2019, with DMSP-based observations through 2013 and simulated VIIRS afterward (6). DMSP records luminance as an integer Digital Number (DN) from 0 to 63, with known saturation near the ceiling (7). Third, we use ACAG surface NO₂ data for the 2005–2019 pollution window (8). Fourth, we use ACAG SatPM PM_{2.5} as a longer-window pollution extension. Fifth, GADM ADM2 boundaries provide the common subnational geography (9).

Treatment is defined at the district-year level. A district is treated when it is the birth region of the national leader in office in that year. To ensure a unique treatment assignment in

Significance Statement

Regional favoritism is a well-known finding in political economy: leaders tend to direct economic activity toward their birth regions. This paper studies how strongly the canonical birth-region nightlights result appears in the later annual windows used for our environmental extension. In those later windows, the reduced form is much weaker than in the long DMSP replication window. The attenuation is broad across geographic, political, and measurement-motivated subsamples. Pollution outcomes do not suggest that leaders shifted from directing more growth to directing cleaner growth toward birth regions.

R.S. conceived the study, built and integrated the main data pipeline, ran the core empirical analysis, and drafted the manuscript. M.M.A. contributed to the NO₂ analysis. R.W. contributed to manuscript review and proofreading.

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transition years, we truncate the earlier of any two consecutive leader spells that overlap within a country.

The core estimating equation uses ADM2 fixed effects and country-year fixed effects:

$$\log(Y_{ict} + 0.01) = \beta \text{BirthRegionLeader}_{ict} + \alpha_i + \gamma_{ct} + \varepsilon_{ict},$$

where i indexes ADM2 regions, c countries, and t years. With ADM2 and country-year fixed effects, β is identified by changes in the current leader's birth district relative to other districts in the same country-year, net of permanent district differences. A causal interpretation requires birth and non-birth districts to have followed parallel within-country trends absent leader ascension. That assumption could fail if future leaders come from districts already on different trajectories or if national transitions coincide with region-specific shocks, so we interpret the estimates as reduced-form evidence rather than a fully causal decomposition.

Standard errors are clustered by leader spell in the baseline, following the leader-term clustering convention in recent replication work such as (3). Because (1) and (4) use country-level clustering, we also report country-clustered robustness checks for the main estimates. Following the baseline specification in (1), we estimate lag 0, lag 1, and lag 2 treatment indicators, and focus on lag 0 in the main text as the most direct test of the regional favoritism hypothesis.

Our later-period environmental outcomes are log nightlights, log NO₂, and a pollution-intensity measure defined as $\log(\text{NO}_2 + 0.01) - \log(\text{Nightlights} + 0.01)$. Because both component regressions use the same regressor and fixed effects, the intensity coefficient is mechanically a linear combination of the NO₂ and nightlights coefficients ($\beta_{\text{intensity}} = \beta_{\text{NO}_2} - \beta_{\text{lights}}$). We report this measure as a compact summary of whether pollution rises more or less than nightlights in birth regions. Because it is constructed directly from the pollution and nightlights outcomes, it is not an independent third test. In the PM_{2.5} extension, the analogous outcome is $\log(\text{PM}_{2.5} + 0.01) - \log(\text{Nightlights} + 0.01)$.

2. Nightlights Reduced Form and Attenuation

Replication in the long DMSP era. The long DMSP replication remains intact. In the replication-style 1992–2013 sample, the lag-0 birth-region coefficient is 0.0128 with $p = 0.030$ (Table 1). This corresponds to a roughly 1.3% increase in nightlights in treated birth regions, in line with magnitudes in the original regional-favoritism literature (1, 10). The same coefficient is larger in 1995–2013 (0.0187, $p = 0.001$) and remains positive, though weaker, in 2000–2013 (0.0110, $p = 0.056$). The baseline activity effect is therefore not a narrow late-1990s artifact.

Attenuation in later windows. The key change is that the nightlights reduced-form point estimate is much smaller and statistically uninformative in later windows. In 2005–2013, still within the DMSP era, the lag-0 coefficient is 0.0017 with $p = 0.768$. In the harmonized 2005–2019 sample, it is 0.0004 with $p = 0.966$. The confidence intervals still contain the historical estimate, so the evidence should be read as a loss of detectable signal rather than proof that the effect is exactly zero.

The window-grid diagnostics in Figure 1 reinforce that reading. Positive coefficients are concentrated in DMSP-only windows with earlier starts and end years around the late 2000s

or early 2010s. Harmonized post-2005 windows, by contrast, remain small and statistically uninformative throughout. The reduced-form problem is not a single bad endpoint choice. It reflects a broader pattern of later-period attenuation.

Additional diagnostics on later-period attenuation. Additional diagnostics narrow the plausible explanations for that attenuation. First, the null is not obviously a pure harmonization artifact. A VIIRS-only check on the standalone 2018–2023 panel is also near zero: the lag-0 coefficient is 0.0038 with $p = 0.722$. Second, the political composition of the sample does not shift enough to make a simple “later years are much more democratic” story convincing. The share of leader-years in the strict-autocracy bin falls only modestly between the long and later windows. The more visible change is shorter observed spell support: average active years per spell fall from 5.22 in 1992–2013 to 3.97 in 2005–2013. This spell-length comparison is partly mechanical because the narrower 2005–2013 window truncates observed active years for spells starting early or ending late.

Third, treated districts in the later DMSP years are already much brighter than untreated districts. In 2005–2013, the median treated district has DMSP luminosity of 16.84, compared with 8.83 for untreated districts, and the treated upper tail sits near the DMSP ceiling at about 61. That is consistent with mechanical attenuation in later years if leaders are increasingly born in already-bright districts. In line with this, the average ascension-year within-country light rank of birth districts is high in both periods and slightly higher in the later sample. Treatment geocoding quality, on the other hand, is stable across periods, so worsening PLAD birthplace measurement is not the leading explanation.

3. Heterogeneity in Attenuation

If the pooled post-2005 activity reduced form masks heterogeneity, the most plausible places to look are settings where the original favoritism mechanism should be strongest or easiest to measure. We therefore test four families of subsamples: geography, political capacity, measurement headroom, and pollution-support samples. The geography cut captures Hodler and Raschky's finding of especially strong effects in Sub-Saharan Africa. The political-capacity cut isolates long-tenure leaders and long-tenure autocrats. The headroom splits ask whether DMSP saturation hides effects in already-bright districts. The pollution-support splits ask whether environmental outcomes are only informative in countries with enough NO₂ variation.

This exercise yields one historical result worth noting but no post-2005 rescue. In the full 1992–2013 DMSP window, the Africa coefficient is 0.0428 ($p = 0.009$) and the Sub-Saharan Africa coefficient is 0.0442 ($p = 0.011$), while the non-SSA coefficient is small and insignificant. The long-run replication is thus concentrated in the geography emphasized by the original literature. In 2005–2013, however, the Africa and SSA estimates turn negative and insignificant. The other theory-favored post-2005 samples in Table 3 also fail to revive the activity reduced form: long-tenure autocrats are consistent with zero and too imprecise to distinguish from the historical effect, dropping high-luminosity districts does not help, and the high-NO₂ country sample remains near zero.

The harmonized 2005–2019 panel gives the most suggestive, though still inconclusive, later pattern. Africa (0.0262,

Table 1. Lag-0 nightlights activity-effect comparison across key windows

Window	Coef.	SE	<i>p</i> -value	N	Treated obs.	Clusters
1992–2013	0.0128	0.0059	0.030	951,915	2,522	2,723
1995–2013	0.0187	0.0055	0.001	829,988	2,196	2,358
2000–2013	0.0110	0.0058	0.056	615,660	1,647	1,759
2005–2013	0.0017	0.0058	0.768	398,636	1,068	1,186
2005–2019	0.0004	0.0087	0.966	671,399	1,701	1,923

All specifications use ADM2 fixed effects, country-year fixed effects, and leader-spell clustering. The 2005–2019 row uses the harmonized DMSP–VIIRS nightlights panel.

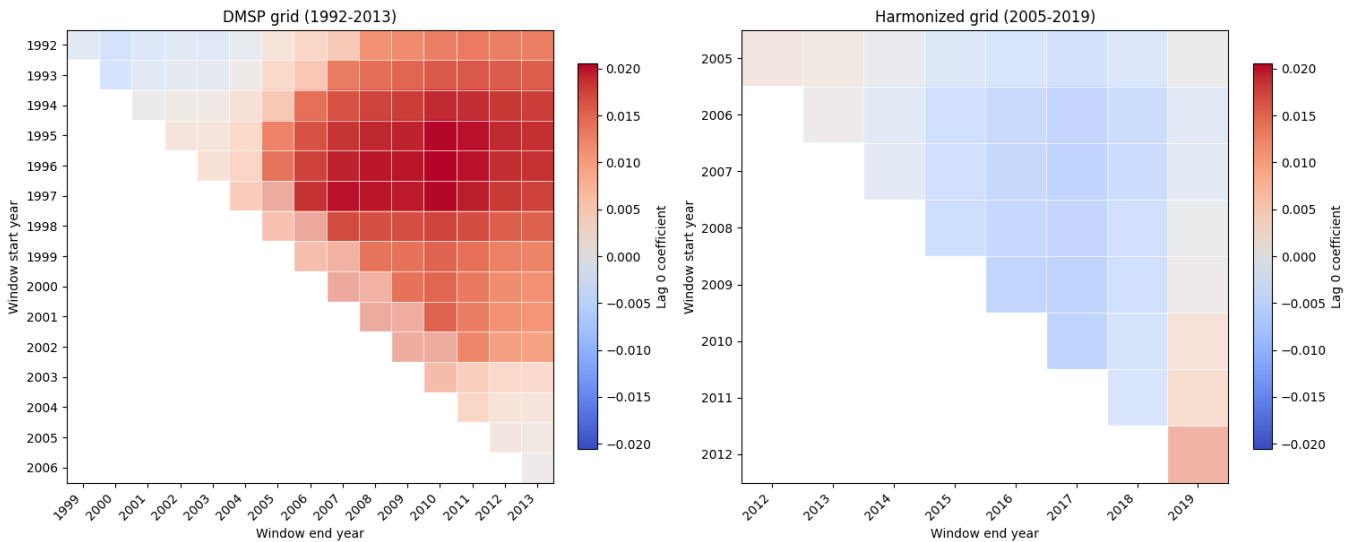


Fig. 1. Lag-0 birth-region nightlights coefficients across valid start and end windows. Each cell is a separate ADM2 fixed-effect and country-year fixed-effect regression; color encodes the birth-region coefficient. Earlier DMSP windows show a block of positive estimates, while post-2005 harmonized windows remain near zero.

Table 2. Robustness to country-level clustering

Spec.	Coef.	Lead. SE	Ctry. SE	Ctry. <i>p</i>
NL, 1992–2013	0.0128	0.0059	0.0075	0.087
NL, 2005–2013	0.0017	0.0058	0.0081	0.832
NL, 2005–2019	0.0004	0.0087	0.0107	0.973
NO ₂ , 2005–2019	0.0308	0.0154	0.0204	0.132
NO ₂ int., 2005–2019	0.0333	0.0179	0.0239	0.163

NL denotes nightlights. All rows use ADM2 fixed effects and country-year fixed effects. Baseline standard errors cluster by leader spell; the robustness column clusters by country, following the original Hodler–Raschky convention.

$p = 0.167$) and Sub-Saharan Africa (0.0315, $p = 0.135$) are larger than the pooled null, but neither clears our minimum-support rule of a positive coefficient above 0.005 with $p < 0.10$. We therefore treat pollution outcomes in these samples as diagnostic rather than confirmatory (Table 4).

4. Pollution-Side Results

Pollution outcomes in the 2005–2019 sample. Pollution estimates test whether the post-2005 attenuation reflects a shift from the quantity to the composition of favored growth. In the pooled 2005–2019 sample (Table 5), the lag-0 nightlights coefficient is near zero, the lag-0 NO₂ coefficient is positive at 0.0308 ($p = 0.045$), and the lag-0 pollution-intensity outcome is also positive at 0.0333 ($p = 0.063$). These point estimates go against a cleaner-growth interpretation. At the same time,

Table 2 shows that the NO₂ estimates become less precise under country clustering, so they are better read as suggestive evidence against green favoritism than as a standalone positive pollution finding.

Autocracy and PM_{2.5} extensions. The heterogeneity checks do not recover the later activity effect either. Restricting the sample to country-years with $v2x_polyarchy < 0.3$ yields a lag-0 nightlights coefficient of 0.0111 ($p = 0.574$) in the DMSP 1992–2013 sample and 0.0076 ($p = 0.552$) in the DMSP 2005–2013 sample. These estimates are positive but too imprecise to rescue the historical result.

The PM_{2.5} extension tells the same story. In longer windows, PM_{2.5} level regressions are positive rather than negative, and in the overlapping 2005–2019 sample the PM_{2.5} pollution-intensity outcome is also not negative. The lag-2 estimate in Table 5 is positive and statistically significant, though we read it cautiously given multiple testing across lags. On balance, pollution outcomes do not explain the weaker post-2005 nightlights result as a shift toward cleaner birth-region growth.

5. Discussion

The main result is that the canonical birth-region nightlights reduced form is much weaker in later periods, especially after 2005. The long DMSP replication still holds, especially in Africa and Sub-Saharan Africa, but the later-period estimates are close to zero and statistically uninformative in the pooled

Table 3. Targeted activity-effect subsample search

Sample	Window	Coef.	SE	p-value	N	Treated
Africa	DMSP 1992–2013	0.0428	0.0163	0.009	109,126	705
Sub-Saharan Africa	DMSP 1992–2013	0.0442	0.0174	0.011	63,378	631
Non-SSA	DMSP 1992–2013	0.0053	0.0059	0.370	888,537	1,891
Pooled	DMSP 2005–2013	0.0017	0.0058	0.768	398,636	1,068
Africa	DMSP 2005–2013	-0.0052	0.0143	0.714	46,939	306
Sub-Saharan Africa	DMSP 2005–2013	-0.0063	0.0164	0.700	28,191	279
Long-tenure leaders	DMSP 2005–2013	-0.0062	0.0087	0.479	300,372	862
Long-tenure autocrats	DMSP 2005–2013	0.0081	0.0173	0.637	42,717	261
Drop top 10% luminosity	DMSP 2005–2013	0.0010	0.0082	0.899	357,745	613
Leader birth DN < 61	DMSP 2005–2013	0.0020	0.0063	0.750	373,862	1,034
Low birth brightness	DMSP 2005–2013	0.0016	0.0108	0.880	142,320	634
High baseline NO ₂ countries	DMSP 2005–2013	-0.0012	0.0069	0.866	312,633	603
Africa	Harmonized 2005–2019	0.0262	0.0189	0.167	83,101	511
Sub-Saharan Africa	Harmonized 2005–2019	0.0315	0.0210	0.135	52,001	466
Low NO ₂ dispersion countries	Harmonized 2005–2019	0.0219	0.0183	0.232	95,961	621

All rows use lag-0 treatment, ADM2 fixed effects, country-year fixed effects, and leader-spell clustering. Long tenure means total PLAD spell length of at least five years. Long-tenure autocracy additionally requires $v2x_polyarchy < 0.3$. The minimum-support rule is $\text{coef.} \geq 0.005$, $p < 0.10$, and at least 50 treated observations in a post-2005 sample. No post-2005 subsample satisfies that rule.

Table 4. Pollution outcomes in strongest positive diagnostic samples

Sample	Activity coef.	Activity p	NO ₂ intensity	NO ₂ p	PM _{2.5} intensity	PM _{2.5} p
Sub-Saharan Africa	0.0315	0.135	-0.0323	0.237	-0.0336	0.127
Africa	0.0262	0.167	-0.0267	0.295	-0.0267	0.178
Low NO ₂ dispersion	0.0219	0.232	0.0156	0.685	-0.0203	0.281

The activity columns use the harmonized 2005–2019 nightlights panel. Pollution-intensity outcomes are log pollution minus log nightlights in the same subsample. These are diagnostic estimates because none of the activity effects is statistically significant under the stated criterion.

sample and in the most natural diagnostic subsamples. This points to a change over time in the strength of the original finding, not just to a change in how nightlights are measured.

The attenuation is broad enough that it cannot be dismissed as a single endpoint choice. We find suggestive patterns in shorter observed leader spells and reduced luminosity headroom, but those explanations are not definitive: the subsample tests dropping high-luminosity districts do not recover the effect, and spell-length attenuation is partly mechanical in the shorter window. The 95% confidence intervals for the post-2005 coefficients in Table 1 still contain the historical 1.3% effect, so we cannot statistically rule out a constant-effect null. The results indicate a loss of detectable signal, not its proven absence.

The pollution analysis narrows one explanation for that attenuation. If post-2005 favoritism shifted from more growth to cleaner growth, pollution intensity should fall in leader birth regions even when nightlights do not rise. Instead, the pooled pollution estimates are zero to positive, and the negative Africa and Sub-Saharan Africa intensity point estimates in Table 4 are too imprecise to carry the cleaner-growth interpretation. The paper therefore does not solve why the reduced form weakens after 2005, but it shows that the weakening is broad and is not explained by a simple shift toward greener birth-region favoritism.

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Data, Materials, and Code Availability

Code, notebooks, and manuscript sources are available in the public GitHub repository at <https://github.com/RichSchulz/political-leader-pollution>. The reported tables are produced from the analysis notebooks in `analysis/`, including `replication.ipynb`, `green_favoritism.ipynb`, `green_favoritism_pm25.ipynb`, `final_subsample_search.ipynb`, and `final_clustering_robustness.ipynb`. External inputs include PLAD, GADM, ACAG, V-Dem, and harmonized nightlights data obtained under the providers' access conditions.

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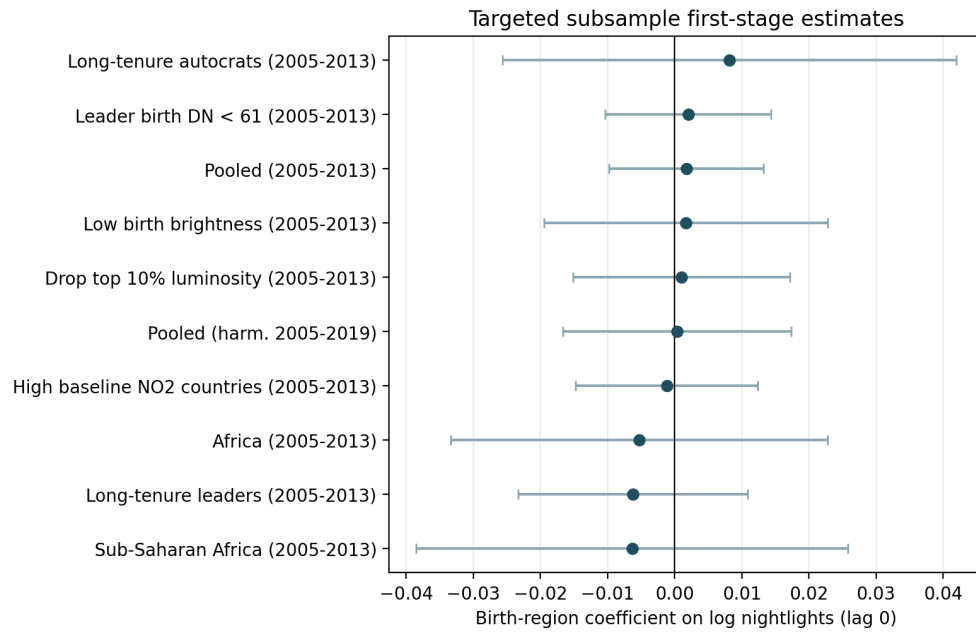


Fig. 2. Targeted post-2005 activity-effect estimates. Points are lag-0 coefficients and horizontal bars are 95% confidence intervals for pooled, Africa, Sub-Saharan Africa, long-tenure, long-tenure-autocracy, luminosity-headroom, low-brightness, and high-NO₂ country samples. The pooled harmonized 2005–2019 estimate is included as a reference row.

Table 5. Main later-period pollution results

Outcome	Window / spec	Coef.	SE	p-value	N
log(Nightlights + 0.01)	2005–2019 lag 0	-0.0026	0.0091	0.778	631,377
log(NO ₂ + 0.01)	2005–2019 lag 0	0.0308	0.0154	0.045	631,377
NO ₂ pollution intensity	2005–2019 lag 0	0.0333	0.0179	0.063	631,377
PM _{2.5} pollution intensity	2005–2019 lag 0	0.0041	0.0095	0.665	669,438
PM _{2.5} pollution intensity	2005–2019 lag 1	0.0122	0.0078	0.120	669,438
PM _{2.5} pollution intensity	2005–2019 lag 2	0.0175	0.0074	0.019	669,438

Pollution intensity is $\log(\text{Pollution} + 0.01) - \log(\text{Nightlights} + 0.01)$. Negative coefficients would indicate cleaner growth in leader birth regions; current estimates are zero to positive.

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